

Unattended Luggage Detection Using a Modified YOLO12 Model

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Abstract—We propose a real-time unattended luggage detection framework based on dual deep learning models, a large-scale public dataset, and explicit spatio-temporal reasoning. We trained a custom YOLOv12m model on a publicly available, manually curated dataset comprising 29,053 images and 130,475 annotated instances across three luggage classes—bag, backpack, and trolley—collected from real surveillance-style videos. To improve detection performance for small and borderline-scale luggage items, we introduce a small-object-aware loss function, while keeping the underlying YOLOv12m architecture unchanged. Compared to the original YOLOv12m, the new model yields consistent performance improvements across all classes, achieving mean gains of 7.1% mAP@0.50, 7.0% mAP@0.50–0.95, and 7.4% F1-score, with the largest improvements observed for small and deformable luggage items.

Keywords—unattended luggage detection, YOLOv12, tracking-by-detection, spatio-temporal reasoning

I. INTRODUCTION

Abandoned luggage detection is a critical problem in video surveillance systems deployed in public environments such as airports, railway stations, and urban areas. The task is particularly challenging due to the small spatial footprint of luggage items, their large visual variability, frequent occlusions, and the need for reliable temporal reasoning to distinguish truly abandoned objects from temporarily unattended ones. Manual monitoring of surveillance footage is inefficient, error-prone, and difficult to scale, motivating the development of automated detection systems. Existing methods can be broadly categorized into background subtraction-based approaches, hybrid computer vision and machine learning systems, depth-based methods, and end-to-end deep learning frameworks.

Early abandoned luggage detection methods relied on background subtraction, motion analysis, and heuristic temporal rules to identify static objects. Several extensions focused on improving static-object detection reliability through trajectory analysis and back-tracing verification [1], while others employed pixel-level state machines and rule-based logic to suppress false positives [2]. While computationally efficient, these approaches were highly sensitive to illumination changes, camera motion, occlusions, and crowded scenes, and lacked semantic understanding of object categories, resulting in high false-alarm rates and missed detections in real-world environments [3, 4].

Several studies explored the use of depth information to improve robustness in complex surveillance scenes. Beleznaï et al. [5] combined stereo depth and intensity cues to detect left luggage, reducing false alarms caused by shadows and illumination changes. Ajami and Lang [6] utilized RGB-D sensors to segment foreground objects and refine ownership estimation based on 3D spatial relationships. Nevertheless,

those methods require specialized hardware and are less practical for deployments using standard monocular systems.

Deep learning-based object detectors have significantly improved robustness in complex surveillance scenarios, enabling reliable detection of people and objects.

Despite these advances, abandoned luggage detection remains challenging. Luggage items often occupy a very small portion of the image, exhibit substantial visual diversity across bags, backpacks, and trolleys, and require consistent association between people and their belongings over time. Abandonment is therefore a spatio-temporal event rather than a frame-level property, making reliable detection particularly difficult in crowded environments.

Recent YOLO-based approaches have demonstrated strong real-time performance for abandoned luggage detection, while requiring careful consideration of scale distribution, class imbalance, and spatio-temporal reasoning. Santad et al. [7] applied YOLO for joint person and luggage detection and used distance thresholds to trigger abandonment alarms. More recently, Vrsalovic et al. [8] proposed a YOLOv11-based framework with dynamic radius estimation, highlighting that static distance thresholds and inconsistent dataset annotations can significantly affect practical performance.

However, most existing systems rely on standard training configurations and loss formulations that are not optimized for small luggage items and are trained on limited or generic datasets that do not adequately represent real-world luggage variability. To address these limitations, we propose a complete abandoned luggage detection framework with the following contributions:

- a) *large-scale public luggage dataset*: we release a publicly available dataset containing 29,731 images and over 130,000 annotated instances across three luggage classes—bag, backpack, and trolley—captured from real surveillance-style videos and augmented to improve robustness [9].
- b) *small-object-aware training for YOLOv12m*: we trained a custom YOLOv12m model using a modified loss function that emphasizes small and borderline-scale luggage items while keeping the network architecture unchanged, achieving clear performance improvements over the original model.
- c) *tracking-based spatio-temporal abandonment reasoning*: we employ YOLOv12x for person detection and associate people and luggage over time using tracking-by-detection. Abandonment is determined through interpretable distance- and duration-based constraints, reducing false alarms caused by temporary separations.

By combining small-object-aware training, dual-model detection, and explicit spatio-temporal reasoning, the proposed system provides an accurate and real-time solution for abandoned luggage detection in real-world surveillance scenarios, addressing key limitations of previous studies.

II. THE PROPOSED MODEL

To address the challenges of realistic unattended luggage detection scenarios, we propose a unified framework built around dual YOLOv12 models, small-object-aware training, and explicit spatio-temporal reasoning. The new model leverages the efficiency and representational power of the YOLOv12 architecture while explicitly targeting the failure modes encountered in surveillance scenarios. In particular, the proposed model introduces two key methodological contributions compared to standard YOLO-based pipelines:

a) *small-object-aware loss function for luggage detection*: Luggage detection is performed using a custom-trained YOLOv12m model optimized through a modified loss formulation that increases the contribution of small and borderline-scale luggage items during training. This strategy improves gradient balance, enhances localization stability, and increases recall for small objects, without altering the underlying network architecture.

b) *spatio-temporal detection framework*: on top of detection and tracking, we introduce a dedicated unattended-detection mechanism that combines distance-based person-luggage association with duration-based temporal constraints. By explicitly modeling abandonment as a spatio-temporal event rather than a frame-level property, this framework reduces false alarms caused by brief separations, occlusions, or transient detector noise, while remaining interpretable and suitable for real-time deployment.

A. Small object-aware loss and assignment

We introduce a modified loss formulation and assignment strategy, based on dynamic curriculum weighting, auxiliary center loss for small objects, adaptive loss clipping, and assignment tuned towards small objects.

To start with, in order to counteract the fact that large bounding boxes tend to dominate the optimization process (because their Intersection over Union (IoU) gradients are stronger), we introduce a dynamic curriculum weighting scheme that combines inverse-area scaling with confidence-based weighting. This design ensures that small objects exert a stronger influence during the early stages of training, preventing them from being ignored. For each positive assignment j , we compute a combined weight:

$$w_j^{(t)} = \alpha^{(t)} \tilde{a}_j + (1 - \alpha^{(t)}) s_j \quad (1)$$

where t is the training epoch, $s_j \in \{0, 1\}$ is the target score, and $\tilde{a}_j$ is a normalized inverse-area term:

$$a_j = \frac{1}{\max(A_j, \varepsilon)}; \quad \tilde{a}_j = \frac{a_j}{\max_{k \in P} a_k} \quad (2)$$

with $A_j = (x_2 - x_1)(y_2 - y_1)$ the ground-truth area (in input pixel coordinates), $\varepsilon > 0$, and P the set of positives in the batch. The mixing coefficient $\alpha^{(t)}$ follows a curriculum:

$$\alpha^{(t)} = \text{clip} \left[0.9 - 0.4 \frac{t}{T}, 0.3, 0.9 \right] \quad (3)$$

using two clipping thresholds set by grid-search, enabling the transition from early area-dominant (favoring small objects) to later balanced learning (T is the total number of training epochs).

We apply to both the Intersection over Union (IoU) and the Distribution Focal Loss (DFL) terms used in the baseline model.

In the second step, in order to compensate for the fact that small objects induce small IoU-based regression losses, (since small shifts cause IoU to collapse, even when centers are close), we added a lightweight L1 penalty on box centers applied only to small objects, using a decaying schedule:

$$L_{center} = \frac{1}{\max(1, \sum_j 1_{small}(j))} \sum_j \|\hat{c}_j - c_j\| \cdot 1_{small}(j) \quad (4)$$

scaled by a decaying weight given by

$$\lambda_{center}^{(t)} = \max \left[0.01, 0.05 \left(1 - \frac{t}{35} \right) \right],$$

where the binary mask 1_{small} isolates the small targets (area lower than 24×24 pixels), c_j and $\hat{c}_j$ are the ground-truth and predicted bounding box centers, respectively.

To continue with, we propose the per-batch clipping of the IoU and DFL losses with epoch-dependent ceilings to alleviate the production of occasionally unstable spikes that may destabilize optimization:

$$L_{IoU} = \min(L_{IoU}, M_{IoU}(t)), \quad M_{IoU}(t) = 10 + 10 \left(1 - \frac{t}{T} \right) \quad (5)$$

$$L_{DFL} = \min(L_{DFL}, M_{DFL}(t)), \quad M_{DFL}(t) = 5 + 5 \left(1 - \frac{t}{T} \right)$$

where the thresholds M_{IoU} and M_{DFL} were set by grid-search.

Finally, in order to prevent false negatives in case of small objects detection, we enlarged the candidate pool dimensionality from $k = 10$ to $k = 25$, which may yield more anchor candidates per bounding box, improving recall results.

The overall detection objective combines these innovations as follows:

$$L^{(t)} = \lambda_{box} L_{IoU} + \lambda_{DFL} L_{DFL} + \lambda_{cls} L_{cls} + \lambda_{center}^{(t)} L_{center} \quad (6)$$

with the λ_{box} , λ_{DFL} , and λ_{cls} weights are set as in the original YOLO12 model, while $\lambda_{center}^{(t)}$ is defined above.

B. Spatio-temporal detection framework

A luggage item must remain outside the vicinity of any person for a sufficiently long duration before it can be considered abandoned. Let $\{I_t\}_{t=1}^T$ denote a video sequence. The goal is to identify luggage objects for which there exists a continuous time interval $[t_s, t_e]$ such that: a) no person is within a spatial radius R of the object during the interval, and b) $t_e - t_s > T_{unattended}$.

To robustly detect object categories, two specialized object detectors based on the YOLOv12 architecture were employed. Person detection is performed using YOLOv12x, selected for its strong performance on large-scale datasets and its high recall and robustness for the “person” class in crowded scenes. Luggage detection is performed using YOLOv12m, trained on a custom luggage dataset containing the classes backpack, bag, and trolley.

Raw luggage detections may include overlapping boxes or inconsistent class labels. To avoid duplicate reasoning, luggage detections are consolidated using class-agnostic Non-Maximum Suppression (NMS) based on IoU. Among

overlapping detections with IoU above 50% only the detection with the highest confidence is retained, regardless of class label. This ensures that each physical luggage item is represented by at most one detection per frame. After consolidation, both person and luggage detections are associated across frames. Each object is represented by a persistent track with a bounding box, last observation time, and estimated velocity. Track motion follows a constant-velocity model.

Detections are matched to existing tracks using a global assignment strategy based on the Hungarian algorithm. For a detection d and a track τ , the association cost is defined as:

$$C(d, \tau) = -w_{IoU} * IoU(b_d, b_\tau^{pred}) + w_{dist} * \|c_d - b_\tau^{pred}\|_2 + \lambda_{cls} \quad (7)$$

where b_τ^{pred} is the predicted bounding box of track τ and λ_{cls} is a soft penalty applied when class labels are inconsistent.

A luggage item is considered supervised if at least one person is within a predefined supervision radius R . The nearest person satisfying this condition is designated as the luggage owner. To avoid rapid owner switching in crowded scenes, ownership changes are accepted only if a new candidate remains significantly closer for a minimum confirmation duration.

For each luggage track, an unattended timer is maintained. By accumulating unattended time only when the luggage is visible and unsupervised, the method avoids false alarms caused by brief separations or detection dropouts.

III. EXPERIMENTAL RESULTS

In this work, we introduce a new large-scale luggage detection dataset, which is publicly available and hosted on the Roboflow platform to ensure transparency and reproducibility [9]. The dataset was curated from approximately 600 publicly available YouTube videos, primarily recorded by fixed or quasi-static surveillance cameras in public spaces, transportation hubs, and indoor environments, addressing the lack of large, realistic surveillance datasets available in the public domain. The dataset is designed for luggage detection and comprises three object categories: backpack, bag, and trolley. In total, 29,053 images containing 130,475 annotated object instances were collected and partitioned into training, validation, and test subsets using a realistic split strategy. The detailed distribution of images and instances across these subsets is reported in Table I.

The resulting per-class distribution—26.7% backpacks, 21.9% bags, and 51.3% trolleys—reflects the natural class imbalance commonly observed in surveillance data, particularly in transportation environments where wheeled luggage is prevalent. This imbalance was intentionally preserved to avoid introducing artificial bias during training. Fig. 1 presents representative dataset samples, illustrating the diversity of scenes, object scales, occlusions, and illumination conditions encountered in the data. Table II summarizes the distribution of object instances across size categories (small, medium, large) under different small (S) and medium (M) normalized area thresholds, computed over the entire dataset. These results highlight the scale-sensitive nature of the dataset and indicate that many luggage instances lie near size boundaries, motivating the use of small-object-aware training strategies to improve detection performance.

TABLE I. DATASET SPLIT AND PER-CLASS INSTANCE DISTRIBUTION

Split	Images	Instances	Backpack	Bag	Trolley
Train	25,302	112,214 (86.0%)	26.8%	22.2%	51%
Valid	2,954	14,371 (11%)	26.9%	20.1%	53%
Test	797	3,890 (3%)	24.1%	22.5%	53.4%
TOTAL	29,053	130,475	26.7%	21.9%	51.3%

TABLE II. OBJECT INSTANCE DISTRIBUTION ACROSS SIZE

Small ($< 24 \times 24$ pixels)	Medium ($[24 \times 24, 64 \times 64]$ pixels)	Large ($> 64 \times 64$ pixels)
16,476 (12.6%)	70,701 (54.2%)	43,298 (33.2%)

Before training, all images were preprocessed to standardize format and improve visual quality, as follows: a) auto-orient: image orientation was corrected using embedded metadata to ensure upright representations and prevent misleading pose variations during training. b) resize: all images were uniformly resized to 640×640 pixels, matching the input requirements of YOLO models. c) auto-adjust contrast: adaptive histogram equalization was applied to enhance visibility and emphasize object boundaries, particularly in low-light or high-glare conditions affecting small objects.

To improve robustness and reduce overfitting, the dataset was expanded using a controlled augmentation pipeline. Each training image was used to generate three independent augmented variants, effectively increasing the training set size by 3 times. Augmentations were applied stochastically using the Roboflow platform, with randomly sampled magnitudes within predefined ranges:

- a) horizontal flip, simulating viewpoint variation
- b) random rotations between -14° and $+14^\circ$, emulating camera tilt or minor motion
- c) horizontal and vertical shearing up to $\pm 13^\circ$, introducing mild geometric distortion
- d) grayscale conversion applied to 10% of images to simulate poor lighting or monochrome camera feeds
- e) Gaussian blur applied with kernel sizes up to 1.6 pixels, replicating motion blur and reduced image quality.

All experiments were performed on a high-performance workstation with an Intel Core i9-13900KF CPU (up to 5.8 GHz) and 64 GB of DDR5 RAM (6000 MHz), enabling efficient data loading and preprocessing for large-scale image datasets. Model training and inference were accelerated using an NVIDIA GeForce RTX 4090 GPU with 24 GB of VRAM, operating with CUDA 12.1.

The experimental pipeline was implemented in Python 3.10.2 using the PyTorch 2.1.2 deep learning framework. The original and the custom YOLOv12m models were trained under identical conditions, using the same dataset splits and hyperparameters.

Table III compares detection performances on the test set using standard object-detection metrics. Across all classes, the modified training configuration consistently outperforms the baseline.

IV. DISCUSSION AND CONCLUSIONS

The results reported in Table III indicate consistent improvements across all object classes and evaluation criteria. The mean mAP@0.50 increases from 0.805 to 0.862, while mAP@0.50–0.95 improves from 0.553 to 0.586, indicating enhanced localization accuracy across a wide range of IoU thresholds. Corresponding improvements in Precision, Recall, and F1-score confirm a more favourable balance between

false positives and false negatives. The most pronounced gains are observed for the bag class, with substantial increases in recall (+14.0%) and F1-score (+11.8%), reflecting a significant reduction in missed detections for small and deformable objects. The backpack class also shows notable improvements, particularly in $mAP@0.50-0.95$ (+7.6%) and recall (+8.0%), suggesting enhanced localization stability across IoU thresholds. For the trolley class, which predominantly contains large and rigid objects, the improvements are more moderate but remain consistent across all metrics, confirming that the training modification does not negatively affect performance on larger targets. Overall, the mean improvements of +7.1% in $mAP@0.50$, +7.0% in $mAP@0.50-0.95$, and +7.4% in F1-score demonstrate a balanced enhancement in detection accuracy and localization quality across object scales.

Class-wise analysis shows that the most significant gains occur for the bag category, which contains a high proportion of small and deformable objects. For this class, notable improvements in recall and F1-score demonstrate increased sensitivity to challenging instances. The backpack category exhibits consistent improvements across all metrics, while the trolley category—dominated by larger and more rigid objects—shows smaller yet stable performance gains.

Overall, the results demonstrate that the proposed training modification effectively enhances detection accuracy,

particularly for small and borderline-scale objects, while maintaining robust performance across all object categories.

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TABLE III. CLASS-WISE DETECTION PERFORMANCES OF THE CUSTOM YOLOV12M MODEL VS THE ORIGINAL YOLOV12M BASELINE

Class	$mAP@0.50$ (Custom)	$mAP@0.50$ (Original)	$mAP@0.50-0.95$ (Custom)	$mAP@0.50-0.95$ (Original)	Precision (Custom)	Precision (Original)	Recall (Custom)	Recall (Original)	F1 score (Custom)	F1 score (Original)
Backpack	0.842	0.801	0.563	0.523	0.821	0.785	0.827	0.766	0.824	0.776
Bag	0.781	0.732	0.502	0.463	0.752	0.687	0.723	0.634	0.737	0.659
Trolley	0.912	0.883	0.643	0.613	0.896	0.840	0.878	0.848	0.887	0.844
All (mean)	0.862	0.805	0.586	0.553	0.823	0.771	0.809	0.749	0.816	0.760

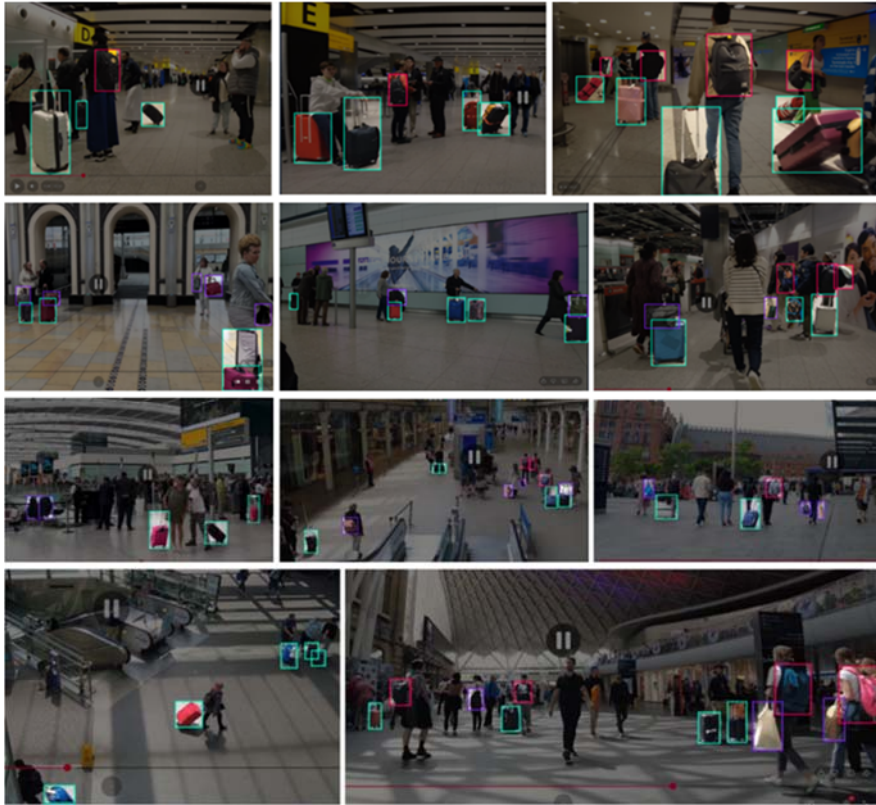


Fig. 1. Sample images from the new luggage dataset.



a)



b)

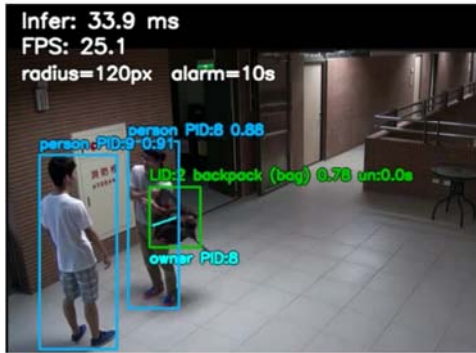


c)



d)

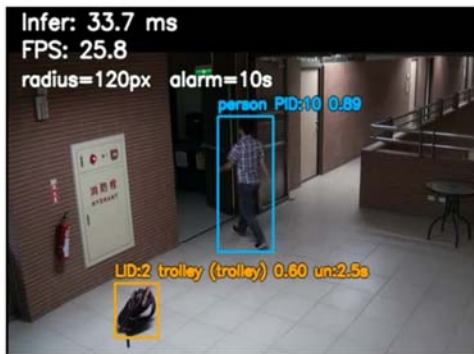
Fig. 2. Unattended luggage detection sequence: (a–b) luggage associated with its owner within a spatial radius; (c) owner exits the radius, activating the unattended timer; (d) alarm triggered after the temporal threshold is exceeded.



a)



b)



c)



d)

Fig. 3. Unattended luggage detection sequence: (a–b) luggage associated with its owner within a spatial radius; (c) owner exits the radius, activating the unattended timer; (d) alarm triggered after the temporal threshold is exceeded